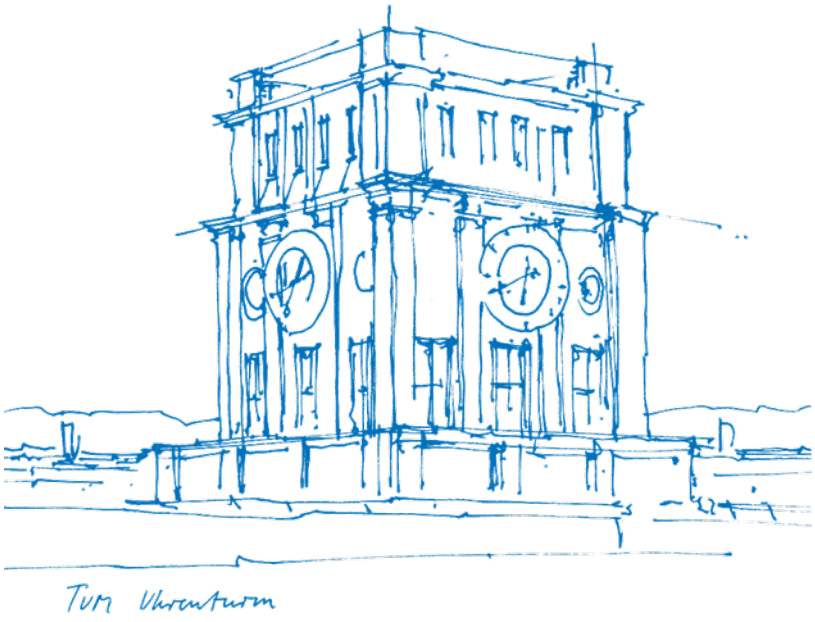
Introduction to Quantum Computing (IN2381)

Prof. Dr. Christian B. Mendl
Technische Universität München
CIT - Department of Computer Science
Quantum Computing
October 14, 2026



Course organization

- **People:**

Christian B. Mendl	lecture
Celina Bräutigam, Xianrui “Ray” Yin	tutorials

- **Moodle:** `https://www.moodle.tum.de/course/view.php?id=121392`
- **Teaching format:** On-campus lectures, 2 on-campus tutorials, 1 online tutorial
- **TUMTutor:** `https://tumtutor.ai/course/40eac6f4-df55-457e-ad2c-b1ff429fd560`
- **Topics:**
  - Mathematical formalism and foundations of quantum mechanics
  - Quantum circuits, quantum teleportation
  - Entanglement, Bell inequalities
  - Quantum search algorithms
  - Density operator
  - Quantum operations

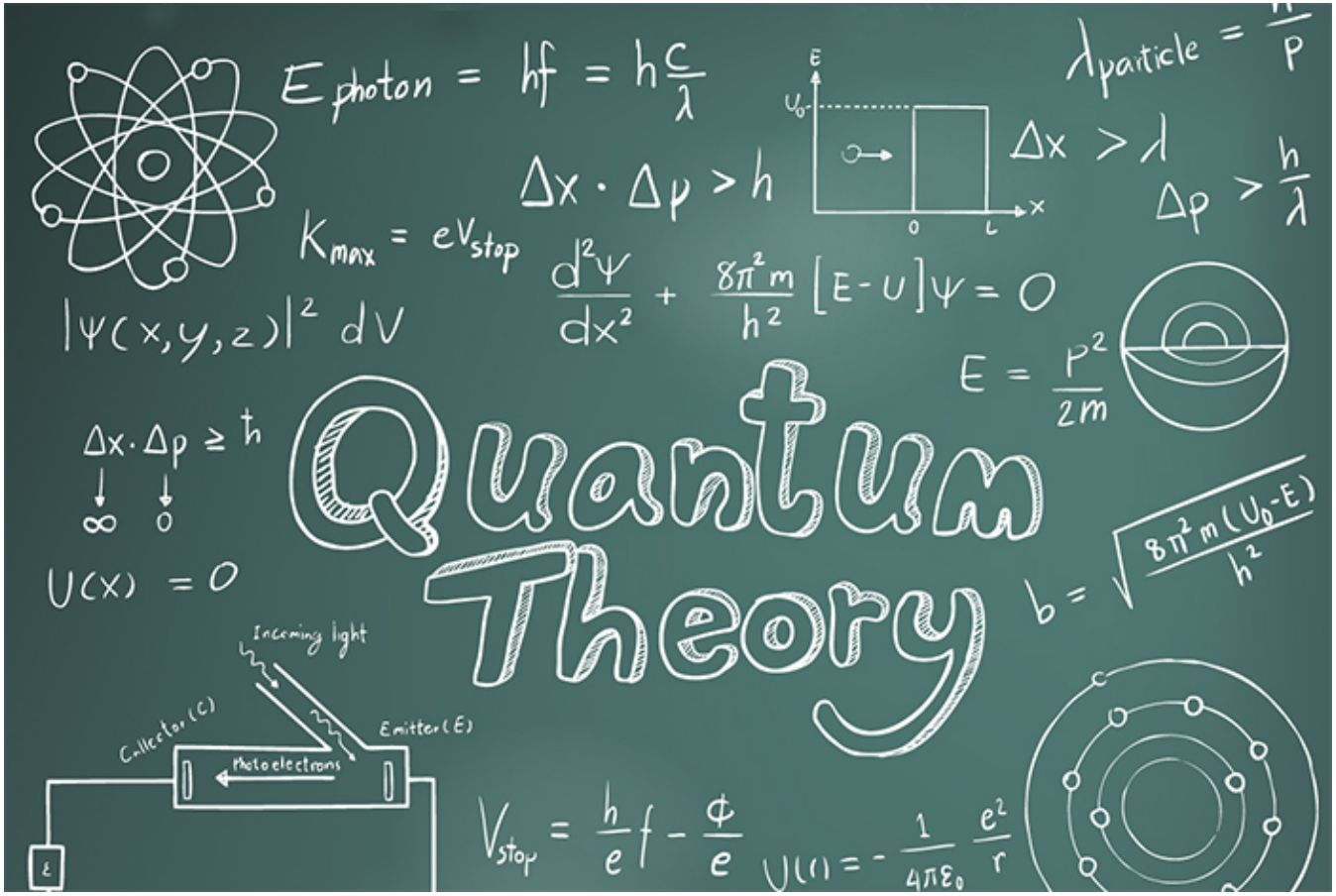


- **Tutorials:** Please attend *one* of the weekly tutorial groups. Content is relevant for exam.
- **Homework:** Try to solve the weekly *voluntary* homework exercises! Reference solutions will be published a week later. No Moodle submission and *no grade bonus* for this instance of the course.
- **Final exam:** On-campus written exam

History of quantum theory and computing

Quantum mechanics

Fundamental physical theory describing behavior of atoms, electrons and other elementary particles
(But: general relativity not taken into account $\rightsquigarrow$ area of active research)



Source: http://timesslate.com/2019/06/21/machine-assimilation-reveals-mysteries-of-quantum-physics/

Quantum theory and computing: a short history (1)

– 1900 Max Planck: energy quantization, invented as mathematical tool to describe black body radiation:

$$E = nhv, \quad n \in \mathbb{N},\ h\text{: Planck's constant},\ v\text{: frequency}$$

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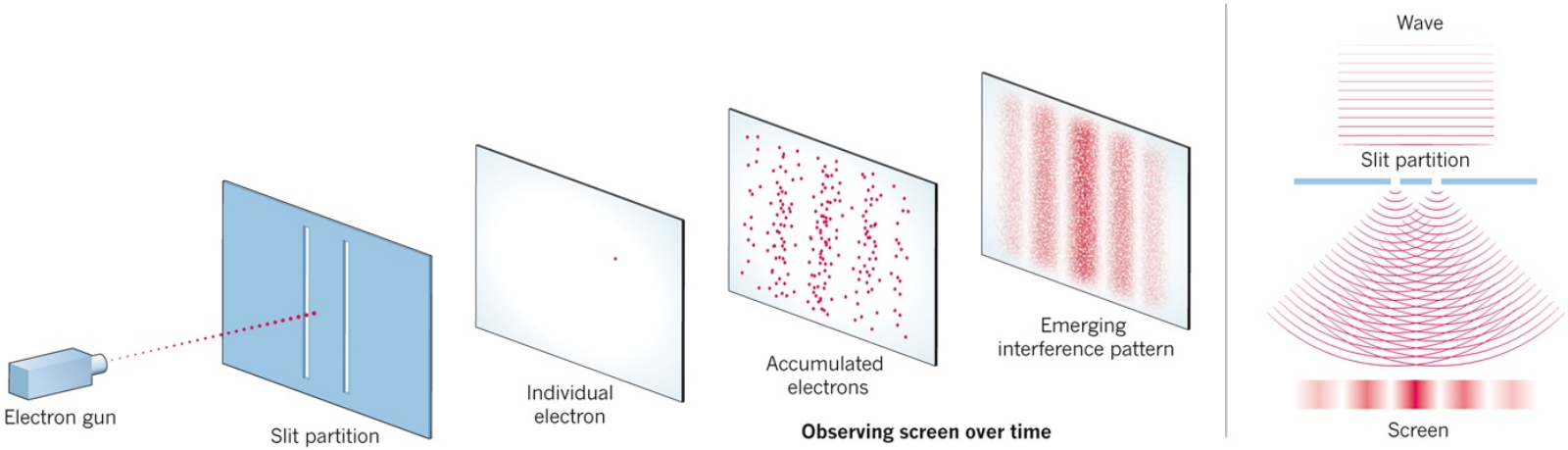
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- 1924 Louis de Broglie: wave–particle duality of matter



Quantum theory and computing: a short history (2)

– 1925 Erwin Schrödinger: Schrödinger equation for the wave function Ψ of a quantum-mechanical system:

$$i\hbar\frac{\mathrm{d}}{\mathrm{d}t}|\Psi(t)\rangle = \hat{H}|\Psi(t)\rangle$$

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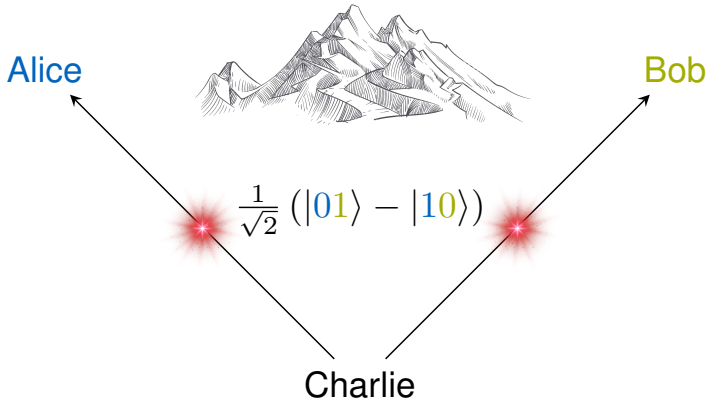
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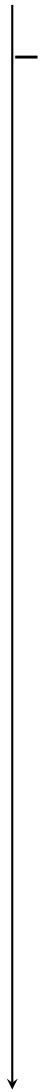
– 1928 Paul Dirac: Dirac equation, combines special relativity and quantum mechanics; first prediction of antimatter

– 1935 Einstein, Podolsky, Rosen: EPR paradox: "spooky action at a distance" due to quantum entanglement
(⇝ Bell's inequality)



Quantum theory and computing: a short history (3)

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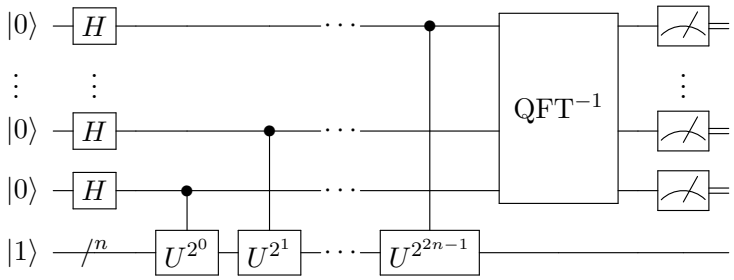
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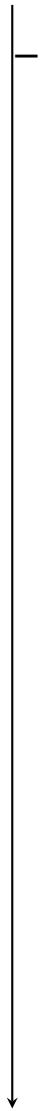
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- 1994 Peter Shor: Shor’s algorithm for integer factorization
  ($\rightsquigarrow$ post-quantum cryptography)



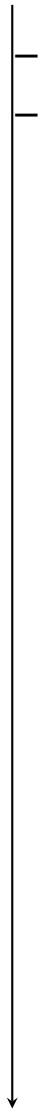
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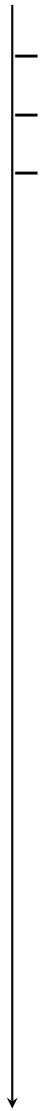
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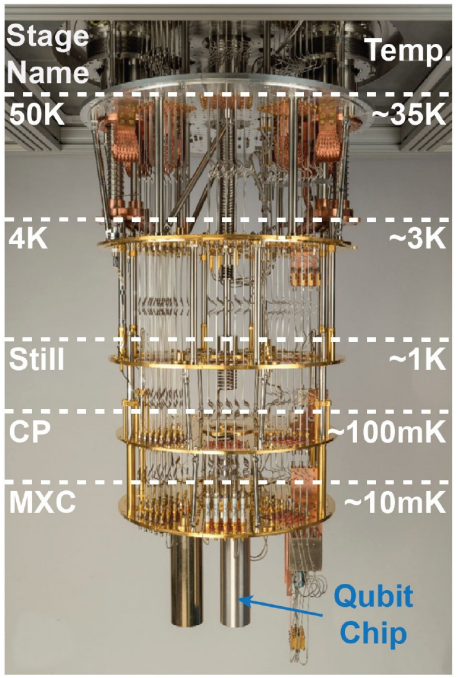
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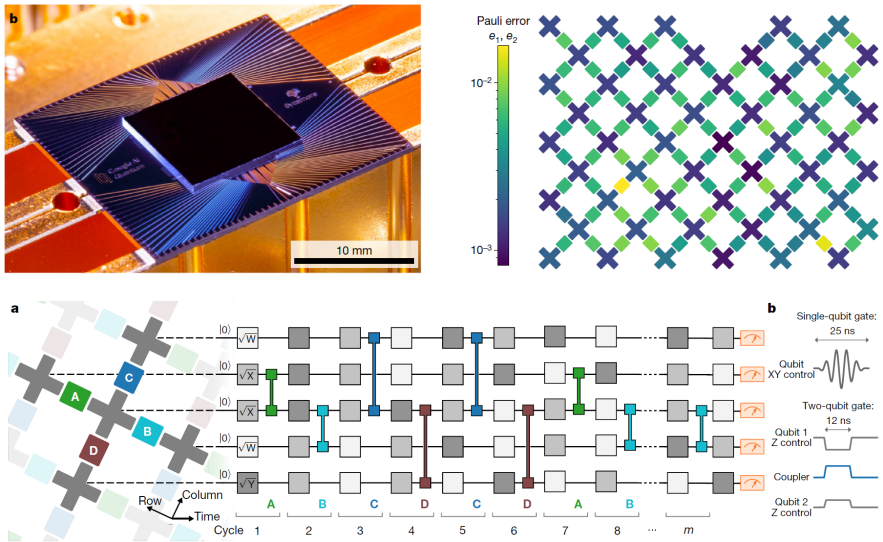
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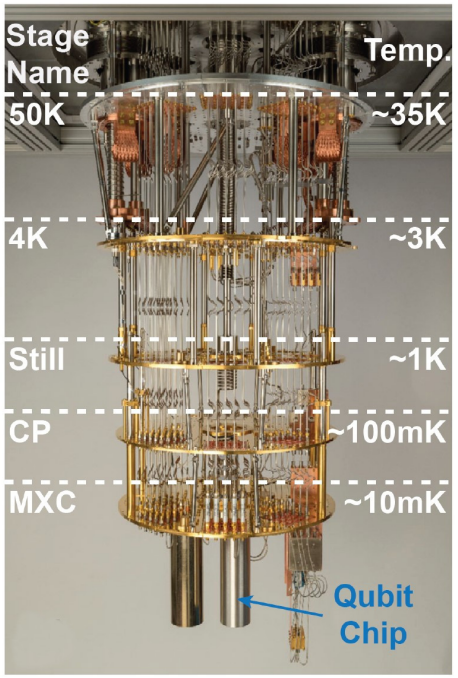
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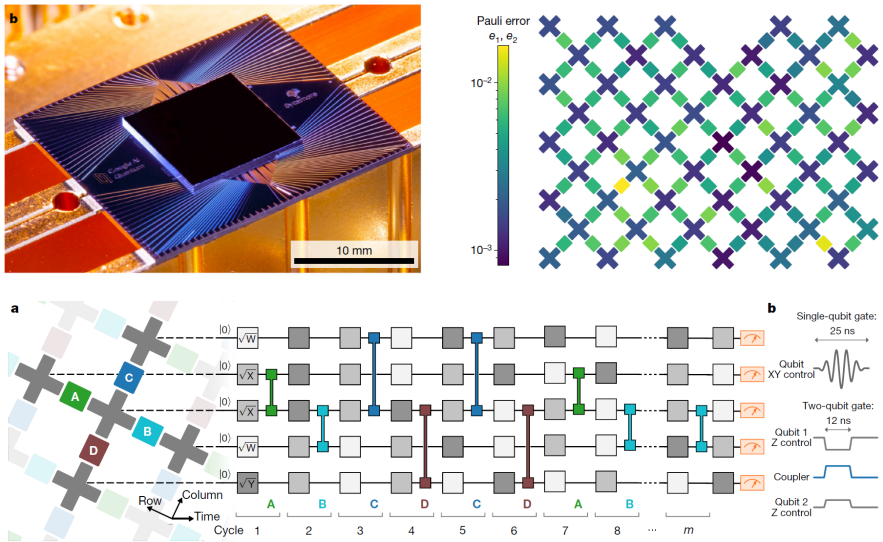


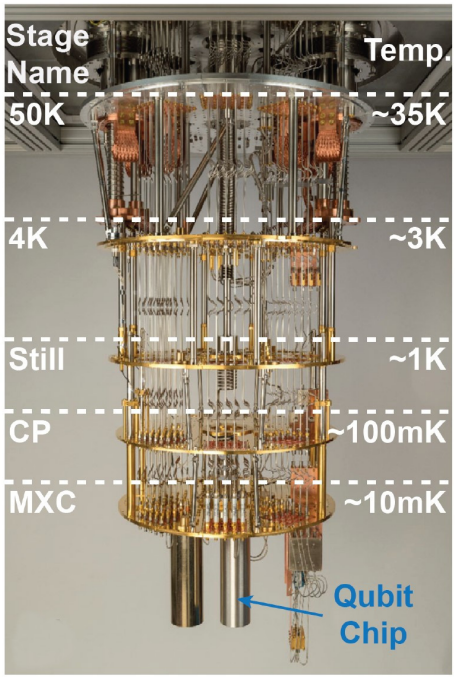


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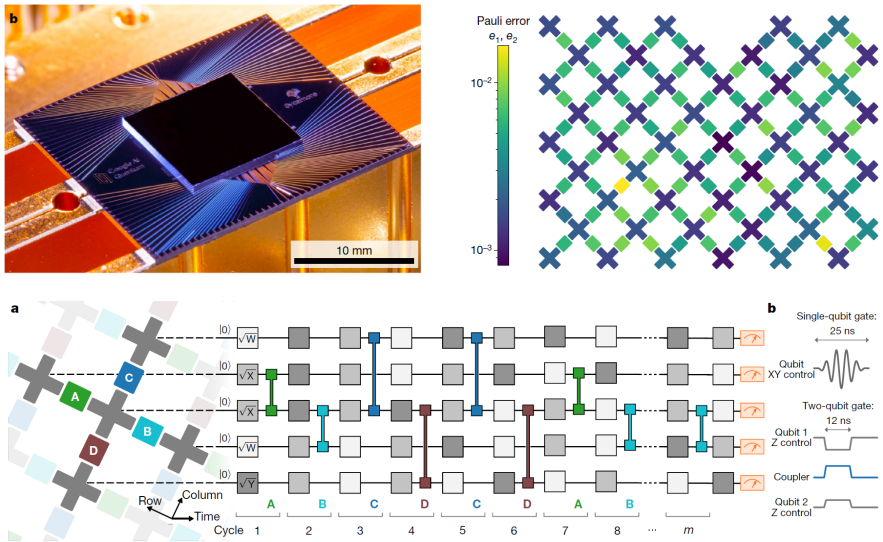


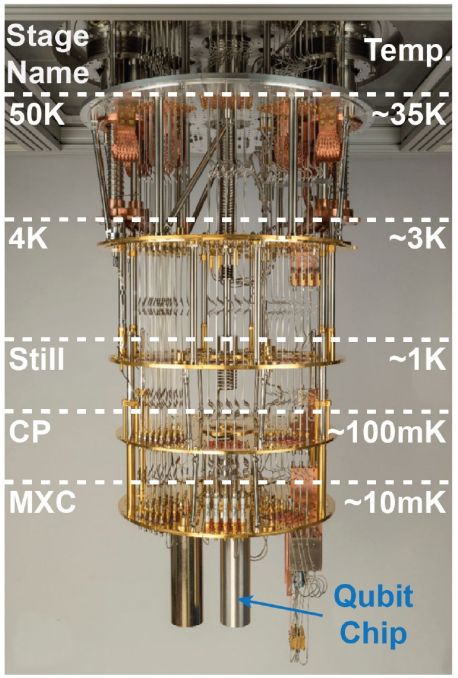
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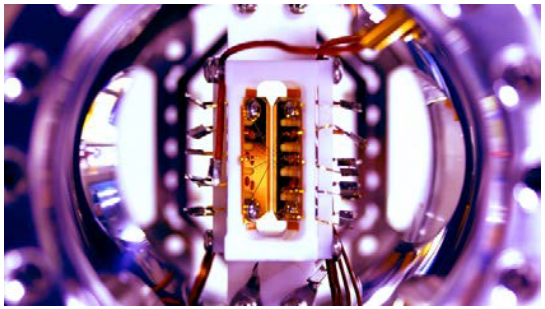




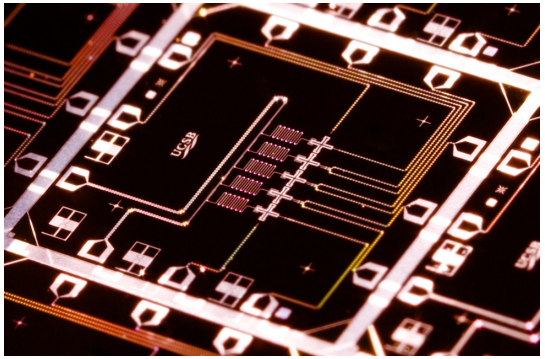
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- 2024 Group of Mikhail Lukin: “Zoned architecture” for ultracold atom arrays

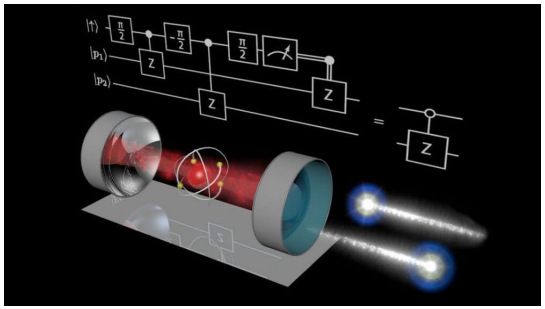
Various approaches for physical realizations



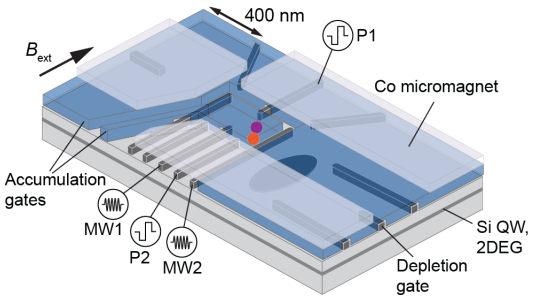
ion traps



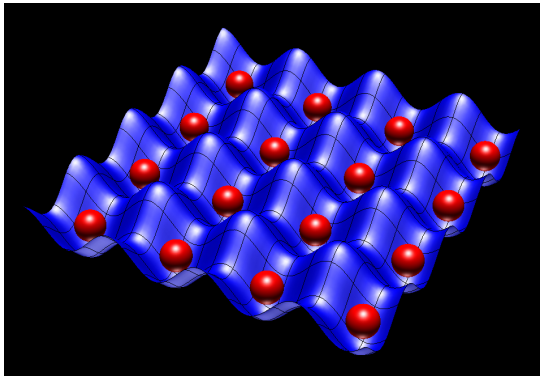
superconducting qubits



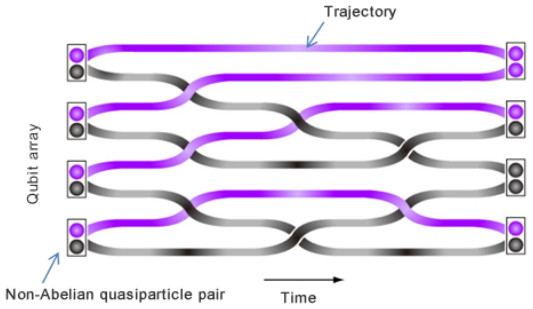
photons



quantum dots



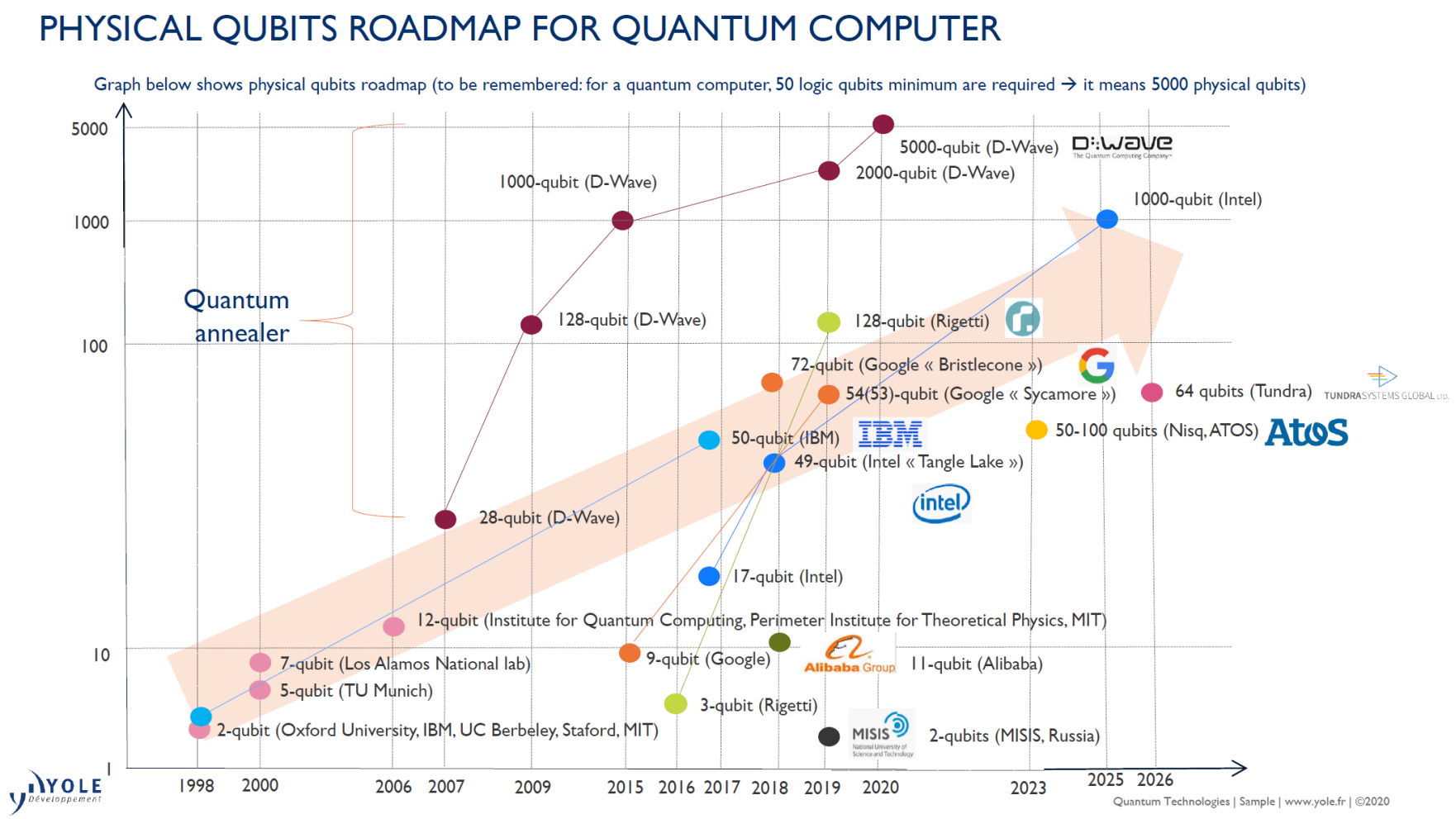
ultracold atoms



topological

Different hardware platforms feasible due to “universality” of quantum mechanics

Physical qubits roadmap



"Qubit quality", gate fidelity, decoherence time, . . . relevant as well

Quantum algorithm examples

Shor's algorithm for integer factorization

$$15 = 3 \cdot 5$$



Peter Shor

Peter Shor (1994)
Quantum algorithm, polynomial w.r.t. number of bits
$\rightsquigarrow$ thread to RSA cryptosystem?

Idea: reduce factoring to order-finding:
given co-prime integers x, N, compute smallest integer $\mathbf{r} \geq 1$ such that

$$x^{\mathbf{r}} = 1 \mod N$$

In case $\mathbf{r}$ is even: find factor of N via

$$x^{\mathbf{r}} - 1 = \left(x^{\mathbf{r}/2} - 1\right)\left(x^{\mathbf{r}/2} + 1\right)$$

Solve order-finding by *quantum phase estimation*

But: not feasible on current quantum hardware: length of circuit, qubit number, low noise tolerance ...

P. W. Shor. "Algorithms for quantum computation: discrete logarithms and factoring" (1994)
C. Gidney and M. Ekerå. "How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits". Quantum 5, 433 (2021)

Shor's algorithm for integer factorization

$$231772171 = ?$$



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$$231772171 = 11279 \cdot 20549$$



Peter Shor

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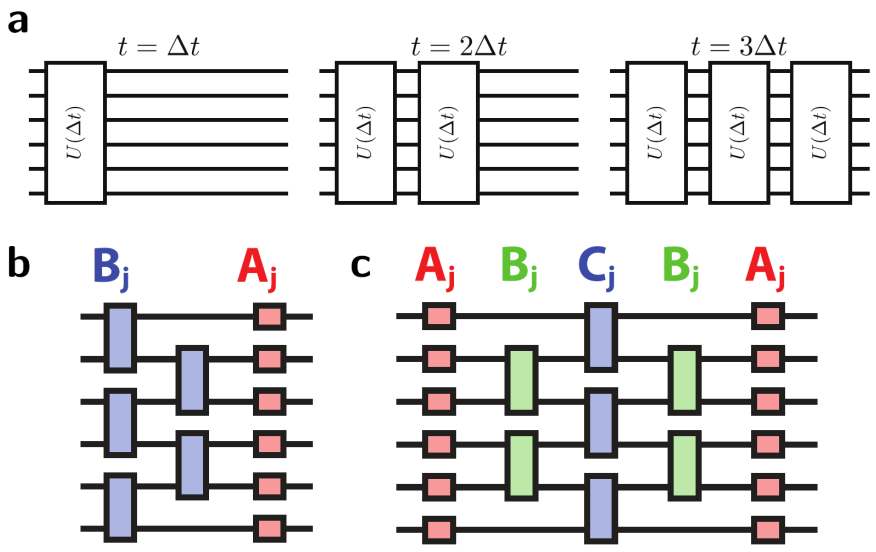
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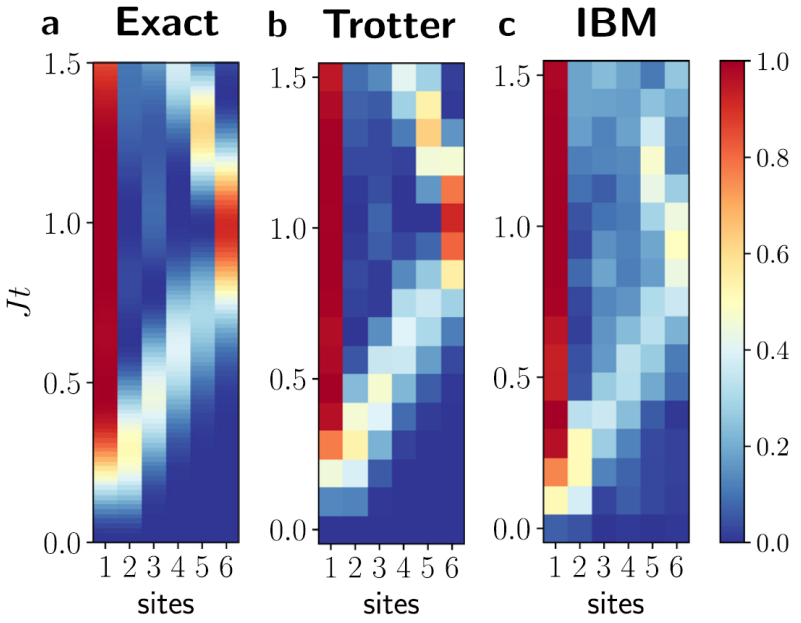
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Simulating quantum many-body dynamics

Approximate e^{-iHt} by Trotter splitting

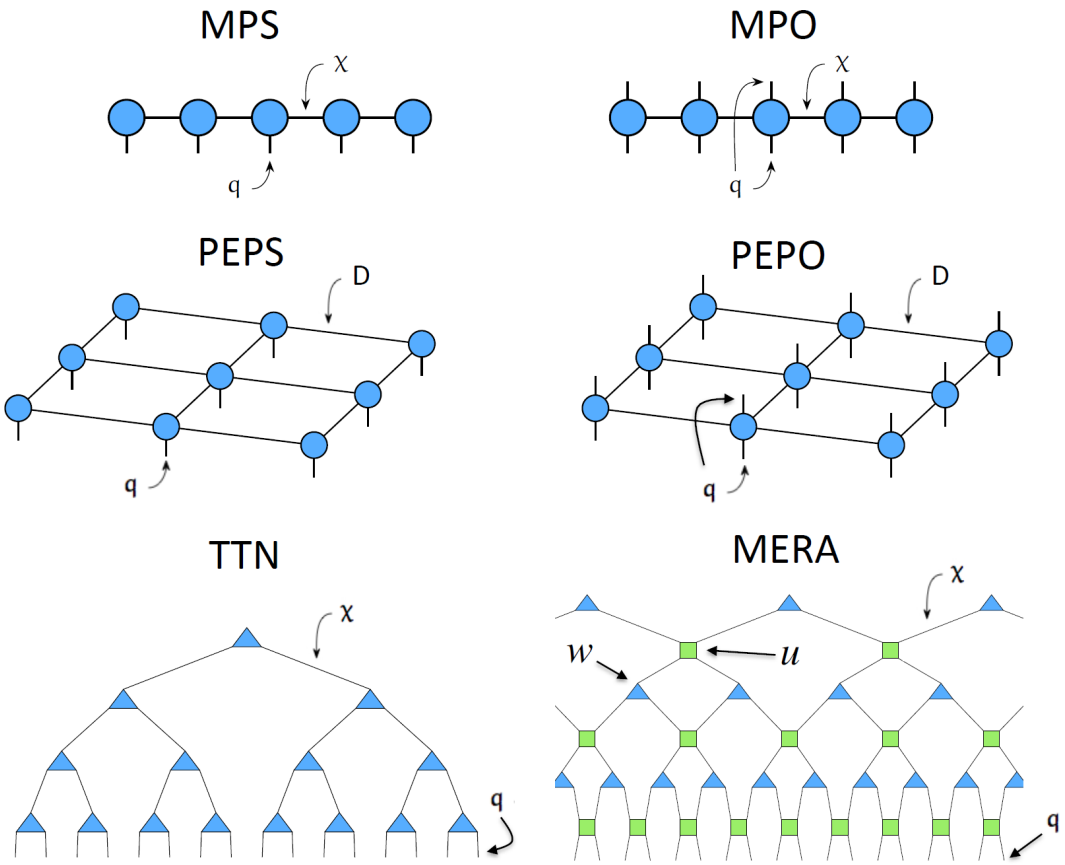




Smith, Kim, Pollmann, Knolle. "Simulating quantum many-body dynamics on a current digital quantum computer". npj Quantum Information 5, 106 (2019)

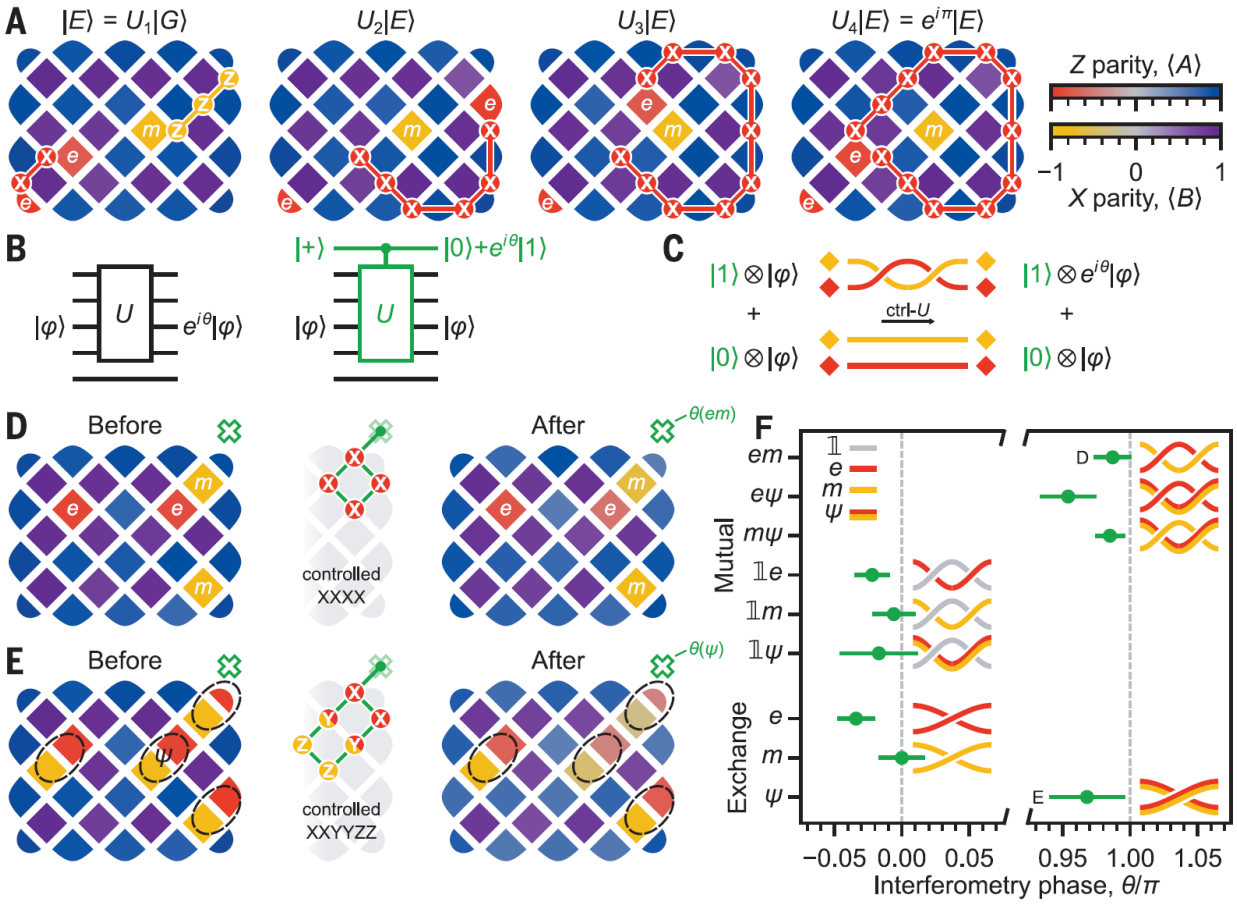
B. Chiaro et al. "Direct measurement of non-local interactions in the many-body localized phase". Phys. Rev. Res. 4, 013148 (2022)

Tensor networks (classical simulation, encoding)



Y. Zhou, E. M. Stoudenmire, X. Waintal. “What limits the simulation of quantum computers?” Phys. Rev. X 10, 041038 (2020)
J. Gray, S. Kourtis. “Hyper-optimized tensor network contraction”. Quantum 5, 410 (2021)
P. Seitz, I. Medina, E. Cruz, Q. Huang, C. B. Mendl. “Simulating quantum circuits using tree tensor networks”. Quantum 7, 964 (2023)

Quantum error correction and topological computing

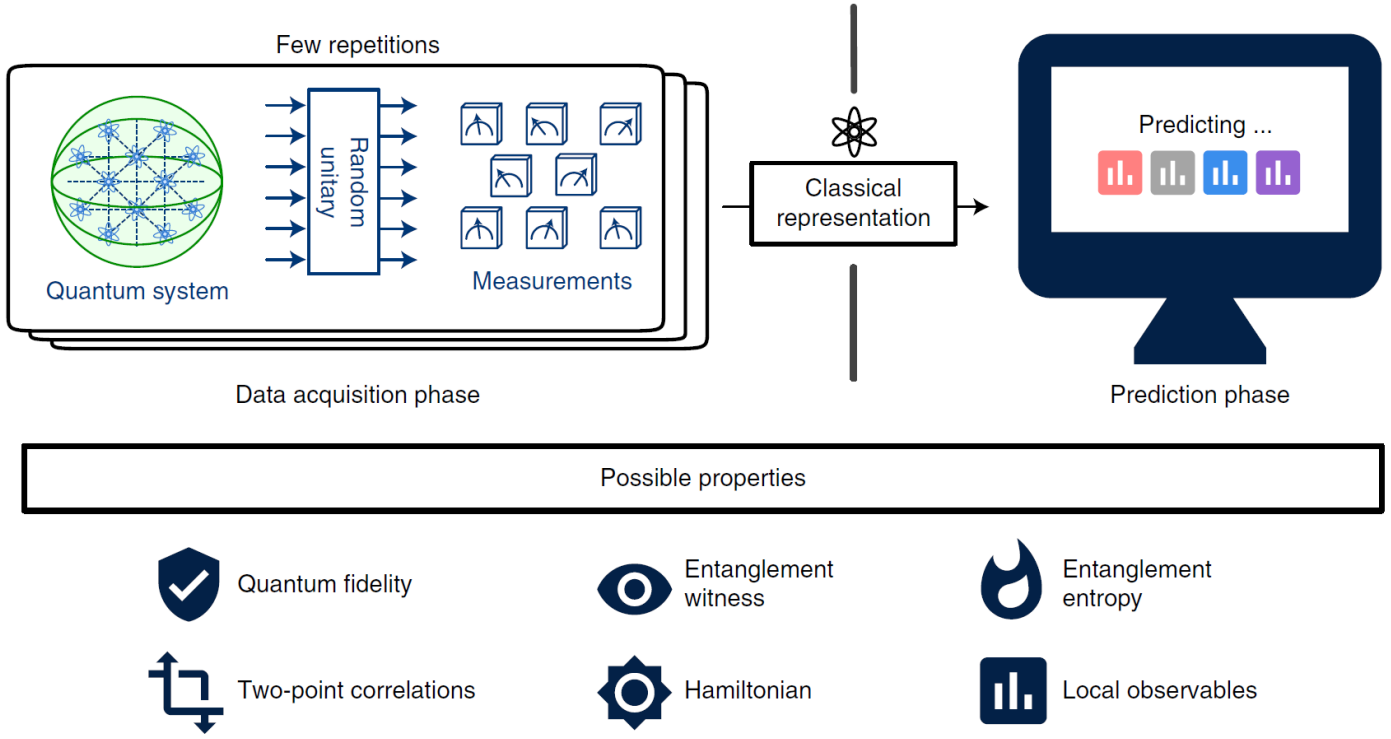


A. Kitaev. “Anyons in an exactly solved model and beyond”. Annals of Physics 321, 2–111 (2006)

E. Campbell, B. Terhal, C. Vuillot. “Roads towards fault-tolerant universal quantum computation”. Nature 549, 172–179 (2017)

K. J. Satzinger et al. “Realizing topologically ordered states on a quantum processor”. Science 374, 1237–1241 (2021)

Randomized measurements and classical shadows



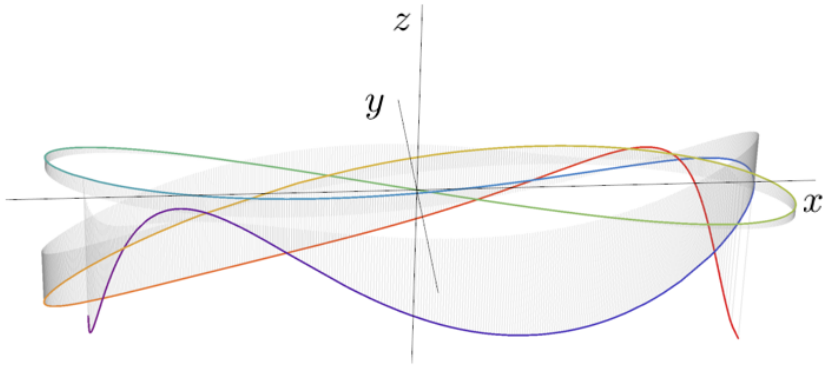
H.-Y. Huang, R. Kueng, J. Preskill. “Predicting many properties of a quantum system from very few measurements”. Nat. Phys. 16, 1050 (2020)

Qubitization, quantum singular value transformation

$$\begin{pmatrix} H & \cdots \\ \vdots & \ddots \end{pmatrix} \xrightarrow[\text{Qubitization}]{e^{\pm i \sin^{-1}[H]}} \xrightarrow[\text{Quantum signal processing}]{} \begin{pmatrix} f[H] & \cdots \\ \vdots & \ddots \end{pmatrix}$$

Input eigenvalue λ

-1 1



Output eigenvalue $\sqrt{1-|\vec{x}_\lambda|^2} + i x_\lambda$

$$U_{\vec{\varphi}} = \mathrm{e}^{i\varphi_0 Z} \prod_{k=1}^{d} W(a)\, \mathrm{e}^{i\varphi_k Z}$$

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